

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming traffic based solely on source IP and port number without maintaining connection state. Which type of firewall is most appropriate? (1 Marks)

- A. Stateful Inspection Firewall
- B. Packet Filtering Firewall
- C. Application-Level Gateway
- D. Next-Generation Firewall

Q2. At which layer of the OSI model does a circuit-level gateway primarily operate to monitor TCP handshaking? (1 Marks)

- A. Network Layer
- B. Transport Layer
- C. Application Layer
- D. Physical Layer

Q3. Why does an Application-Level Gateway (Proxy Firewall) typically experience more latency than a Packet Filtering Firewall? (1 Marks)

- A. It operates at the physical layer.
- B. It must establish two separate connections and inspect full payload data.
- C. It only inspects the IP header.
- D. It uses hardware-based ASIC chips for routing.

Q4. How does a stateful inspection firewall determine if an incoming packet is part of an existing, authorized session? (1 Marks)

- A. By checking a dynamic state table of active connections.
- B. By decrypting the application payload.
- C. By performing a reverse DNS lookup on every packet.
- D. By blocking all UDP traffic by default.

Q5. Which feature distinguishes a Next-Generation Firewall (NGFW) from a traditional stateful firewall? (1 Marks)

- A. Simple packet filtering by port.
- B. Deep packet inspection and application-level awareness.
- C. Operating strictly at Layer 2.
- D. Lack of support for VPNs.

Q6. A company needs to protect its web servers specifically against SQL injection and Cross-Site Scripting (XSS) attacks. Which specialized firewall should be deployed? (1

Marks)

- A. Packet Filtering Firewall
- B. Circuit-Level Gateway
- C. Web Application Firewall (WAF)
- D. Stateful Inspection Firewall

Q7. What is a primary advantage of a host-based firewall compared to a network-based firewall? (1 Marks)

- A. It protects the entire network segment simultaneously.
- B. It can inspect encrypted traffic before it is decrypted by the OS.
- C. It provides tailored protection for a specific device based on its local context.
- D. It requires no local system resources to run.

Q8. In a highly virtualized cloud environment, which type of firewall is best suited to monitor east-west traffic between virtual machines on the same physical host? (1 Marks)

- A. Hardware-based perimeter firewall
- B. Virtual Firewall (vFirewall)
- C. Circuit-level gateway
- D. Stateless packet filter

Q9. If a stateless firewall rule allows outbound TCP traffic on port 80, what additional rule is required for the return traffic to be allowed back to the client? (1 Marks)

- A. No rule is needed because return traffic is automatically tracked.
- B. An inbound rule allowing TCP traffic from source port 80.
- C. A rule blocking all UDP traffic.
- D. A rule enabling deep packet inspection.

Q10. In a classic three-homed firewall architecture, where should publicly accessible servers (like web and mail servers) be placed? (1 Marks)

- A. Inside the trusted internal network.
- B. Directly on the public Internet.
- C. Within a Demilitarized Zone (DMZ).
- D. Behind a loopback interface.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Packet filtering firewalls inspect packets individually based on header information like IP addresses and ports without tracking state.

Q2. Correct Answer: B

Rationale: Circuit-level gateways operate at the Session/Transport layer to verify TCP handshakes and establish sessions.

Q3. Correct Answer: B

Rationale: Proxy firewalls act as intermediaries, establishing separate connections for client and server while inspecting deep application data.

Q4. Correct Answer: A

Rationale: Stateful firewalls track active connections in a state table to validate incoming packets against established sessions.

Q5. Correct Answer: B

Rationale: NGFWs combine traditional firewall capabilities with deep packet inspection, intrusion prevention, and application awareness.

Q6. Correct Answer: C

Rationale: A WAF is designed specifically to analyze HTTP/HTTPS traffic at the application layer to block web-based exploits.

Q7. Correct Answer: C

Rationale: Host-based firewalls run locally, allowing customized security rules tailored to the specific host's applications and services.

Q8. Correct Answer: B

Rationale: Virtual firewalls run as software appliances within virtualized environments to inspect traffic between virtual machines.

Q9. Correct Answer: B

Rationale: Stateless firewalls do not track connection state, requiring explicit rules for both outbound and inbound traffic directions.

Q10. Correct Answer: C

Rationale: A DMZ isolates publicly accessible services from the internal network, limiting exposure if a public-facing server is compromised.